Weekly Meeting 4

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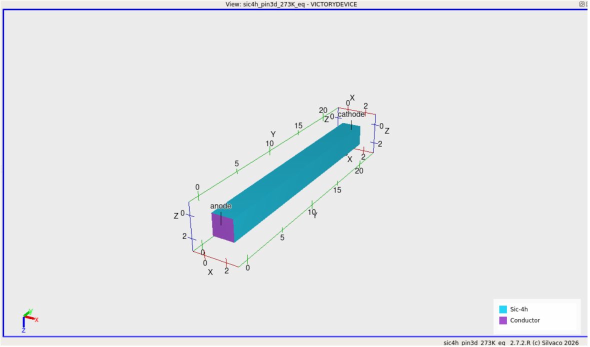
This week's path

Repeated 4H-SiC PIN breakdown attempts → Separated field criterion from visible current → Moved to edge control: no JTE vs multi-zone JTE → Tested a single-event transient at 1000 V

Repeated 4H-SiC PIN breakdown attempts	Separated field criterion from visible current	Moved to edge control: no JTE vs multi-zone JTE	Tested a single-event transient at 1000 V
7–8 deck variants	ionization integral + seeded IV	geometry and electric field	current versus time

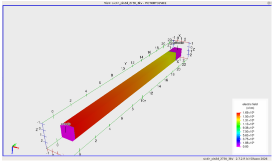
The common question is: which part of the simulated response is physical, and which part is only a numerical diagnostic?

Closest 4H-SiC PIN baseline



3D laterally uniform PIN: $2 \times 2\ \mu\text{m}^2$, total thickness $22.3\ \mu$m.

- $0.30\,\mu$m Al-doped p^+ anode.
- $20\,\mu$m N-doped drift region at $10^{15}\,\text{cm}^{-3}$.
- $2\,\mu$m N-doped n^+ cathode.
- Temperature: 273.15 K.
- Fine mesh at the p^+/n^- junction.



At 3 kV, the plotted peak field is about 1.69 MV/cm.

Model 1: incomplete ionization

$$N_A^- = \frac{N_A}{1 + g_A \exp\left(\frac{E_A - E_F}{k_B T}\right)}$$

$$N_D^+ = \frac{N_D}{1 + g_D \exp\left(\frac{E_F - E_D}{k_B T}\right)}$$

- Al acceptors: $g_A = 4$, $E_A \approx E_V + 0.20$ eV.
- Nitrogen donors: $g_D = 2$.
- Lower temperature reduces the active Al fraction further.
- Enabled with `INC.ION`.

Why it matters: active space charge sets the depletion field, ionization integral and predicted breakdown voltage.

Victory Device incomplete-ionization model; 4H-SiC dopant parameters.

Models 2–3: Fermi statistics and BGN

Fermi–Dirac statistics

$$f(E) = \frac{1}{1 + \exp[(E - E_F)/(k_B T)]}$$

- Used in the $10^{19}\,\mathrm{cm}^{-3}$ contact layers.
- Avoids the non-degenerate Boltzmann approximation there.

Band-gap narrowing

$$E_g^{\mathrm{eff}} = E_g - \Delta E_g(N)$$

- Enabled as `BGN` in the PIN deck.
- It is a high-doping correction near the contacts.
- No custom BGN coefficients were fitted.

These are supporting contact-region models; they are not the reason for the avalanche turn-on.

Models 4–5: low- and high-field mobility

Low-field mobility

$$\mu(N) = \mu_{\min} + \frac{\mu_{\max} - \mu_{\min}}{1 + (N/N_{\text{ref}})^{\alpha}}$$

- Closest PIN deck: `ANALYTIC`.
- JTE and SEU decks currently use `CONMOB`.

High-field saturation

$$v(E) = \frac{\mu_0 E}{[1 + (\mu_0 E/v_{\text{sat}})^{\beta}]^{1/\beta}}$$

- Enabled with `FLDMOB`.
- Prevents drift velocity increasing linearly without limit.

Current limitation: `CONMOB` must be checked against a 4H-SiC-specific mobility parameter set before quantitative comparison.

Models 6–7: SRH and Auger recombination

Shockley–Read–Hall

$$U_{\mathrm{SRH}} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}$$

- PIN deck: $\tau_n = \tau_p = 10^{-7}$ s.
- Supplies depletion-region generation/recombination.

Auger

$$U_A = (C_n n + C_p p)(np - n_i^2)$$

- Relevant mainly at high carrier concentration.
- Included in the seeded PIN deck.
- Default coefficients are used.

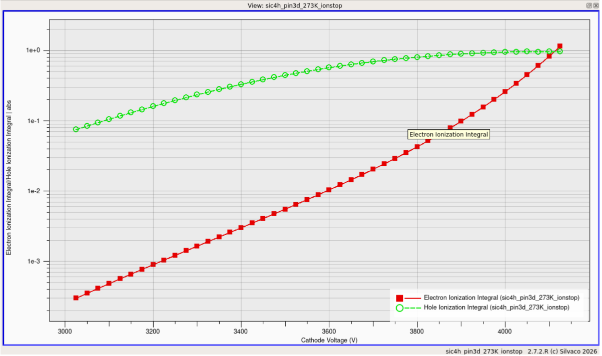
Neither model creates a calibrated dark-leakage prediction without validated lifetime and defect parameters.

Model 8: anisotropic impact ionization

$$\alpha(E) = A \exp\left[-\left(\frac{B}{E}\right)^{\beta}\right]$$

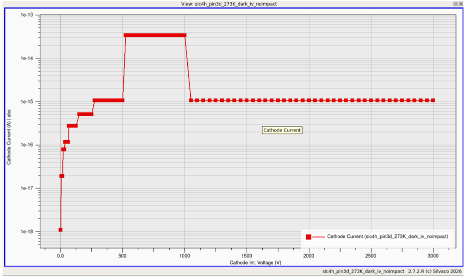
$$I_{e,h} = \int \alpha_{e,h}(E)\,\mathrm{d}s$$

- `SIC4H0001`, anisotropic 4H-SiC model.
- Physical criterion: $\max(I_e, I_h) = 1$.
- No large terminal current is required for this criterion.

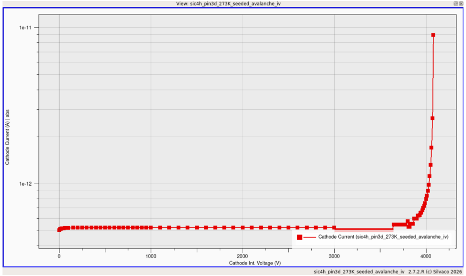


Ionization-integral stop scan. The two carrier integrals reach unity differently.

What the closest PIN run actually showed



Dark, impact-off current: extremely small and affected by sweep-state changes.



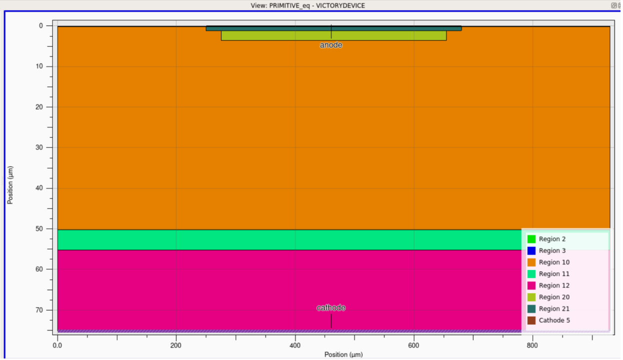
Weak 0.30 μm optical seed, $B_1 = 10^{-4}$ W/cm^2, impact enabled.

Not suitable for extracting a clean dark breakdown curve.

Shows visible multiplication near 4 kV, but it is photo-assisted IV.

Conclusion: use the ionization integral for the breakdown estimate; use the seeded IV only to demonstrate current multiplication.

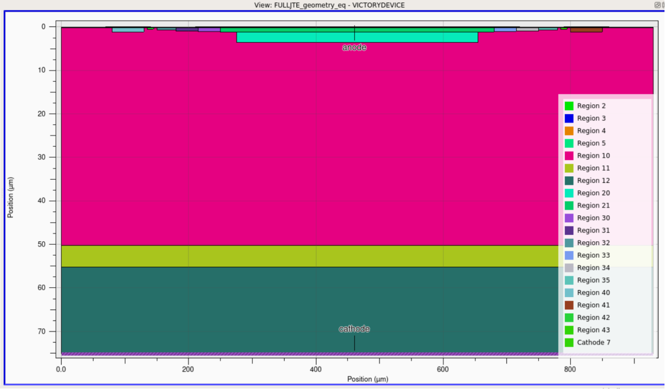
Why the next step was edge termination



Primitive LGAD geometry: active pad and gain layer, no JTE, guard ring or field-plate overhang.

- A vertical PIN baseline does not test lateral edge fields.
- The primitive LGAD creates an abrupt end to the active junction.
- That edge can reach the critical field before the central active area.
- This gives a controlled "bad geometry" reference.

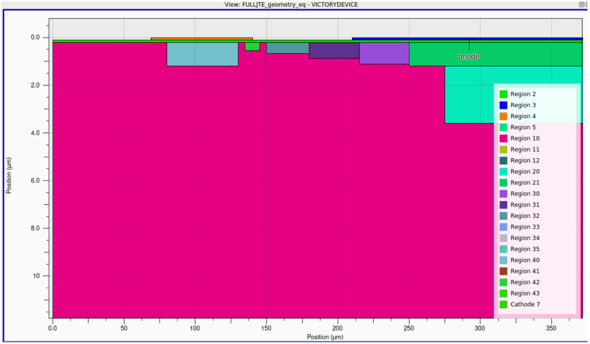
Full LGAD termination structure



Full-width region view; geometry is symmetric about the active pad.

- Central p^{++} active implant.
- Narrower n-type gain layer below it.
- Three p-type JTE zones on each side.
- Outer p-type guard rings.
- Oxide passivation and Al field plate.
- n^{-} epi, n^{+} buffer and n^{++} substrate.

How the multi-zone JTE is intended to work



Left-side termination zoom: guard ring, p-stop, JTE3, JTE2, JTE1 and active region.

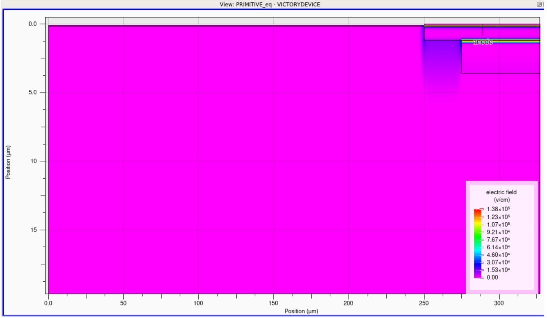
Concentrations in cm^{-3}. These are engineering starting values, not optimized parameters.

$$\nabla \cdot (\varepsilon \nabla V) = -\rho$$

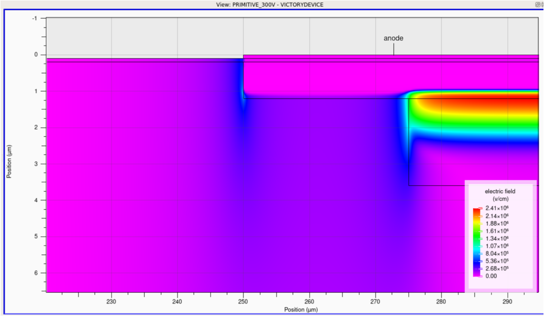
Zone	Width	Al concentration
Inner JTE	$35\,\mu$m	8×10^{16}
Middle JTE	$35\,\mu$m	3×10^{16}
Outer JTE	$30\,\mu$m	1×10^{16}

- Grading spreads depletion laterally.
- The field plate extends $40\,\mu$m beyond the active pad.
- Guard ring gives the termination a controlled outer end.

Primitive LGAD: equilibrium to 300 V



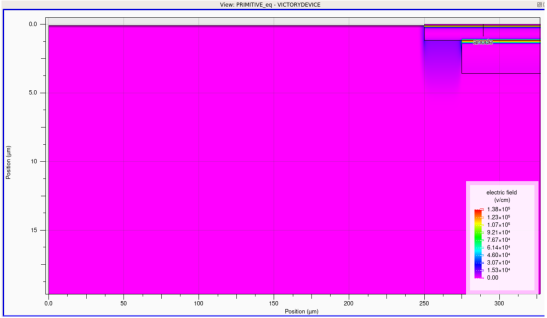
Equilibrium: built-in junction field, plotted scale to 1.38×10^5 V/cm.



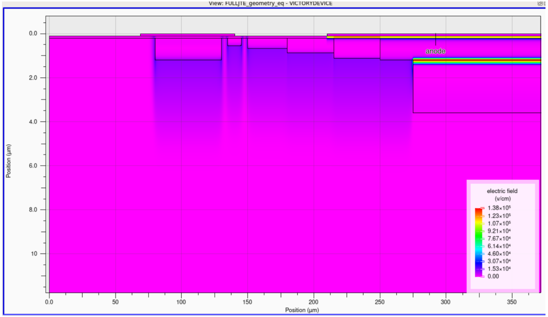
300 V: peak field concentrates near the active/gain-layer edge.

The 300 V map reaches about 2.41 MV/cm at the edge, which is the behaviour the termination is meant to control.

Matched equilibrium comparison



No JTE, equilibrium.

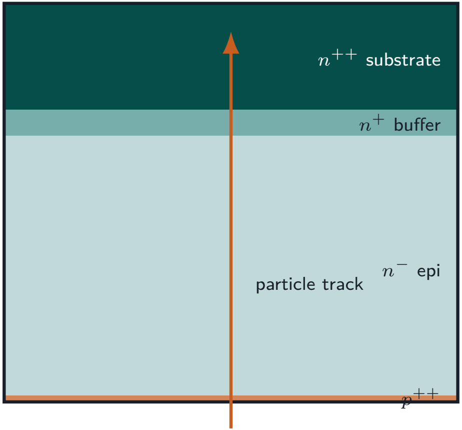


Multi-zone JTE, equilibrium. Same 1.38×10^5 V/cm color range.

What this comparison supports: the added implants reshape the built-in field along the surface.

What it does not yet support: improved high-voltage breakdown; that requires a matched-bias JTE field map and a parameter sweep.

Single-event transient setup



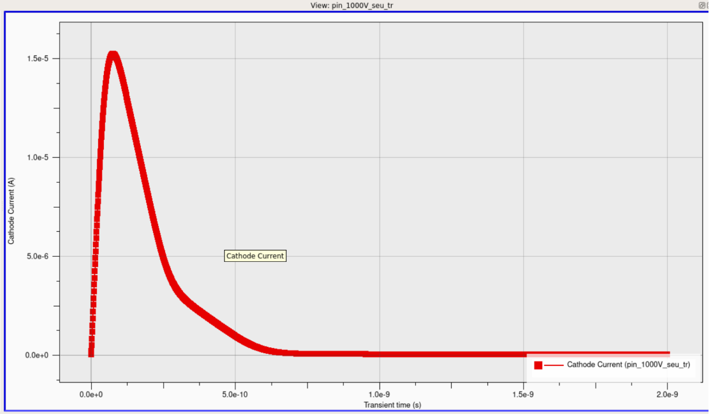
$$G(\mathbf{r}, t) = S_{\text{track}}(\mathbf{r})\, T(t)$$

- Bias: 1000 V; impact ionization off.
- Track: top to bottom at $x = 150\,\mu$m.
- Radius: $0.20\,\mu$m.
- Density: $10^{15}\,\mathrm{cm}^{-3}$.
- $t_0 = 10$ ps, $t_c = 50$ ps.
- `SEU.INTEGRATE` integrates the narrow track over the mesh cells.

This is a toy single-event source. The deposited charge has not yet been calibrated to a MIP or LET.

Victory Device Single Event Effects model: spatial dependence multiplied by time dependence.

Transient cathode-current response



- Prompt rise after the event begins.
- Peak current is about 1.5×10^{-5} A.
- The current then falls as generated carriers are collected.
- Response returns close to baseline within roughly 0.7 ns.

$$Q_{\mathrm{col}} = \int I(t)\,dt$$

Next quantitative check: integrate the pulse and compare collected charge with the charge inserted by the source model.

Next steps

1. Clean the 4H-SiC PIN breakdown deck and separate dark, seeded and ionization-integral outputs.
2. Verify a 4H-SiC-specific mobility model before using the JTE or SEU results quantitatively.
3. Produce matched-bias no-JTE/JTE field maps, then sweep JTE dose and width.
4. Integrate the transient current and calibrate the single-event density to a physical charge or LET.
5. Continue looking for ONSEMI geometry and implant parameters for a realistic device target.

Immediate priority: one matched, explainable comparison before adding more geometry.